

Sugandh Kumar, Ph.D.

Innovative genomic medicine researcher and computational, AI biologist
Senior Computational Biologist UCSF | Computational Genomics & AI/ML | Chemoinformatics

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Professional Summar

More than 10 years of experienced in biomedical and bioinformatic research scientist with a strong academic and industry background including five years of postdoctoral research experience at UCSF. Proven track record of designing scalable analytical pipelines, discovering clinical biomarkers, and driving precision medicine workflows. Expert in high-resolution functional genomics, translational bioinformatics, and machine learning architectures applied to complex autoimmune, inflammatory, and oncological diseases. Deep technical expertise in single-cell multi-omics integration, spatial transcriptomics, multi-ethnic GWAS, and polygenic risk scoring to map pathological cell states and therapeutic pathways. Exceptional record of peer-reviewed scholarship featuring over 30 publications, including multiple first and co-corresponding authors in high-impact manuscripts under active review or accepted. Highly collaborative and motivated to spearhead groundbreaking discoveries, with extensive experience coordinating cross-functionally across NIH-funded projects, pharmaceutical partnerships, and multidisciplinary clinical teams.

Core Competences

Single-Cell & Spatial Genomics

scRNA-seq | CITE-seq | scATAC-seq | Xenium Spatial | Visium/ST | Cell Annotation | Pseudotime | RNA velocity | Cell chat| GSEA | IPA

GWAS & Polygenic Risk Scoring

Multi-ethnic GWAS | Regression | PRS calculation & validation | MIS/TOPMed imputation | PLINK2 | vcfTools | bcftools | GATK | Axiom genotyping arrays

Bulk Genomics & Epigenomics

Bulk RNA-seq (STAR, DESeq2, edgeR) | ChIP-seq | ATAC-seq | WGS/WES (GATK, ANNOVAR) | Methylation | WGCNA co-expression | Networks

Machine Learning & AI

Classical ML | Generated AI | CNN | PyTorch | TensorFlow | Feature selection | Ensemble modeling | Biomarker discovery | Treatment response prediction

Programming & Infrastructure

R | Python | Bash | Perl | HTML | CSS | JavaScript | HPC/HPC clusters | AWS | Docker | Snakemake | Git/GitHub | Linux

Chemoinformatics & Drug Discovery

Molecular Modeling | Docking | MD simulations | QSAR/ADMET | Schrodinger Suite | Automation Tools | ROSETTA

Work Experience

Postdoctoral Research Fellow

June 2021 -- Present

University of California, San Francisco (UCSF) -- Department of Dermatology, Liao Lab

Conducted advanced single-cell multi-omics integration and computational analysis of diverse datasets—including scRNA-seq, CITE-seq, scATAC-seq, Xenium Spatial Transcriptomics, bulk RNA-seq, ChIP-seq, WGS, and GWAS—from tissue biopsies and PBMCs to resolve immune cell infiltration pathways in chronic inflammatory skin and musculoskeletal diseases (Psoriasis, Atopic Dermatitis, Psoriatic Arthritis, and Spondyloarthritis).

Leveraged pharmacogenetics and machine learning architectures to predict patient clinical responses to biologic therapies (including anti-IL-4/IL-13 and anti-IL-23 pathways) by integrating longitudinal scRNA transcriptomic, clinical, laboratory, and genomic data from Phase III clinical trials.

Developed AI/ML-driven biomarker discovery pipelines using scRNA-seq/CITE-seq expressions and GWAS data to discover personalized therapeutic targets, aiming to improve patient outcomes and optimize clinical trial stratification between Psoriasis (PsO) and Psoriatic Arthritis (PsA) in translational dermatology.

Investigating the immunological impact of sleep architecture on chronic systemic inflammation by tracking longitudinal cell-state shifts via custom CITE-seq tracking from PBMC workflows.

Led a multi-ethnic Genome-Wide Association Study (GWAS) cohort analysis, identifying novel risk variants and elucidating complex genotype-phenotype associations to define the genetic architecture of Hidradenitis Suppurativa (HS), psoriasis, atopic dermatitis.

Architected proprietary, in-house polygenic risk score (PRS) computation engines using genetic variant arrays to quantify the cumulative statistical impact of risk alleles on patient disease susceptibility.

Directed a multimodal AI engineering project to develop a web-based clinical tool utilizing computer vision on sequential images and video feeds; automated real-time Psoriasis Area and Severity Index (PASI) scoring to objectively track longitudinal treatment efficacy.

Collaborated cross-functionally with multiple principal investigators and industry partners, ensuring timely delivery of milestones, and contributed key preliminary data to successful NIH and pharmaceutical research grants.

Provided academic leadership and mentorship to undergraduate researchers and clinical medical fellows, supervising technical bioinformatics workflows and experimental execution.

Research Associate2015 -- 2021

Institute of Life Sciences, Bhubaneswar, India

Cancer Biomarkers: Developed ML-based biomarkers for early-stage head and neck squamous cell carcinoma (HNSCC) using microRNA expression profiling; published in PeerJ (2020).

Multi-Omics Cancer Genomics: Conducted comprehensive genomic analyses for oral cancer including bulk RNA-seq, ChIP-seq, GATK variant calling, methylation, and metagenomic datasets.

Computational Drug Discovery: Developed an automated virtual screening pipeline for large chemical compound libraries; performed molecular docking, MD simulations, and QSAR modeling for kinase receptors and other cancer target genes for small molecules design against oral and head and neck cancer.

Comprehensive HNSCC Database: Developed a comprehensive data based for the head and neck cancer genomics expression, 4k full length model structures, millions of synthetic and natural compounds, mutational impact, etc.

Supervised and mentorship: Supervised and provided mentorship to several under Ph.D. research fellows with several Master of Sciences (M.Sc.,) fellow on computational drug designing and ML model used in biology

Collaboration initiation: Cross-functional bioinformatics collaborations with multiple principal investigators internally and across external institutions, driving computational workflows that culminated in several peer-reviewed publications.

Software Trainee (Internship)2014 -- 2015

Apt Software Pvt. Ltd., Kolkata, India

- Proactively delivered structural and computational expertise to influence efforts in targets and lead identification as well as lead optimization
- Developed QSAR models for predicting HIV drug activity; contributed to software testing and scripting.
- Worked on complex data analysis through R and python where analysis situation decided further molecular validation.
- Worked closely with inter-disciplinary projects teams deployed in multiple therapeutic area

Education

Ph.D., Computational Biology2015 -- 2020

Institute of Life Sciences, Bhubaneswar, India |
Junior & Senior Research Fellowship, University Grants Commission (UGC)

M.Sc., Bioinformatics2011 -- 2013

Central University of South Bihar, India | Merit Scholarship

B.Sc., Biotechnology2008 -- 2011

Punjab Technical University, India

Selected Publications

First-Author & Corresponding Author (Under Review / In Press)

1. Jacqueline M. Frost*, Sugandh Kumar* et al., Genetic Analysis of Pediatric Psoriasis. [Accepted: JID, 2026 -- Co-first Author]

2. Sugandh Kumar et al., Dupilumab Reprograms Cutaneous Immune, Neuronal, and Stromal Cells in Atopic Dermatitis. [\[Under Review: JACI, 2025 -- 1st Author/Co-corresponding\]](#)
3. Sugandh Kumar et al., Cellular Indexing of Transcriptomes and Epitopes (CITE-Seq) in Hidradenitis Suppurativa identifies dysregulated cell types and facilitates ML-based diagnosis. [\[Under Review: Nature Communications, 2025 -- 1st Author/Co-corresponding\]](#)

Published Peer-Reviewed Articles (Selected)

4. Faye Orcales, **Sugandh Kumar** et al., A partitioned polygenic risk score reveals distinct contributions to psoriasis clinical phenotypes across a multi-racial cohort. Journal of Translational Medicine, 2024.
5. Bahram Razani, **Sugandh Kumar** et al., A20 (Tnfrsf3) in keratinocytes restricts an early antiviral gene program associated with psoriatic skin and joint disease. JID, 2024.
6. Gian Carlo, **Sugandh Kumar** et al., Immunological differences in Atopic Dermatitis across age groups: Insights from single-cell multi-omics. Journal of Translational Medicine, 2026 (Accepted).
7. Antonia Wiala, **Sugandh Kumar** et al., Patients with Psoriasis and Suppurative Hidradenitis (PSOSH) share genetic risk factors. JAAD, 2025.
8. Jared Liu, **Sugandh Kumar** et al., combining single-cell transcriptome and surface epitope profiling with ML to diagnose psoriatic arthritis. Frontiers in Immunology, 2022.
9. Audrey Bui, **Sugandh Kumar** et al., A weighted genetic risk score using 88 susceptibility variants in Newfoundland psoriasis patients. Frontiers in Genetics, 2023.
10. **Sugandh Kumar**, Srinivas Patnaik, Anshuman Dixit, Predictive models for stage and risk classification in HNSCC. PeerJ, 2020.

Full list of publication : [Google Scholar](#): *More than 30 peer-reviewed articles, 5 book chapter author, 1 Australian innovation patent (No. 2021103325). Active reviewer 200+ manuscript of 30+ journals, journals editors across 6 journals.*

Awards, Recognition & Service

- Selected for SID world leadership in Dermatology and Immunology Conference - 2026
- NIH Grant Update Lecture, UCSF Immunology Conference (invited speaker) - 2022
- Best Poster Award -- Genomics India 2019 International Conference in Human Genomics
- Junior & Senior Research Fellowship, University Grants Commission (UGC), Government of India (2015-2020)
- Journal Editor: Frontiers in Bioinformatics (2023); Guest Editor: MDPI Biomolecules (2023)
- Active reviewer: Scientific Reports, Cell Reports, Frontiers in Pharmacology, MDPI Journals (100+ reviews)
- Australian Innovation Patent No. 2021103325 -- Method to Detect Paternal Contributors in Seminal Extracellular Vesicles for Recurrent Pregnancy Loss

~ References and other documents available upon request

Cover letter

June 3, 2026

Hiring Committee

Kenai Therapeutics

San Francisco Bay Area, CA

Re: **Senior Scientist, Bioinformatics**

Dear Hiring Committee,

I am writing to apply for the **Senior Scientist, Bioinformatics** role at Kenai Therapeutics. Kenai's mission -- advancing next-generation allogeneic cell therapies for neurological diseases -- sits at exactly the intersection where I have spent the past decade building expertise: translating high-resolution single-cell and multiomics data into actionable insights that drive therapeutic decisions. I hold a **Ph.D. in Computational Biology** and have completed five years of postdoctoral research at **UCSF**, where I have designed, built, and operated production-grade genomics pipelines, applied ML to Phase III clinical trial data, and contributed to three first-author manuscripts currently under review at **Nature Communications, JACI, and JID**. I am confident that my technical depth, translational mindset, and comfort operating in small, fast-moving teams make me a strong match for this role.

Single-Cell and Spatial Multi-Omics

The role calls for deep hands-on experience with the full analytical lifecycle of single-cell and spatial omics data - from raw data through biological interpretation. This is the core of my UCSF work. I have built and operated end-to-end pipelines across **scRNA-seq, CITE-seq (198 surface proteins + transcriptome), scATAC-seq, and Xenium In Situ Spatial Transcriptomics** from human tissue biopsies and PBMCs across >200 samples. I am proficient in **Seurat, Scanpy, Squidpy, and scVI** -- all three frameworks listed in your requirements -- and have extensive experience with cell type annotation, trajectory analysis, pseudotime, RNA velocity, cell-ligand interaction (CellChat), and GSEA.

Two highlights most relevant to allogeneic cell therapy:

- Led a paired pre/post dupilumab intervention study (n=24 patients, scRNA-seq + Xenium Spatial Transcriptomics) that resolved how IL-4Ra blockade reshapes neuronal, stromal, and immune compartments in human skin at single-cell resolution -- identifying novel MPZ+S100B+ Schwann cell populations and CD44 as mechanistic hubs. Under review: JACI 2025 (first and co-corresponding author).
- Built a CITE-seq pipeline integrating 198 surface proteins with transcriptomes across >100 PBMC samples from patients with HS, Psoriasis, PsA, and Spondyloarthritis. Developed ML classifiers (Random Forest, XGBoost) achieving ~87% diagnostic accuracy for disease stratification. Under review: Nature Communications 2025 (first and co-corresponding author).

Machine Learning, Statistics, and AI-Accelerated Workflows

Your requirements emphasize strong foundations in statistics and ML, proficiency with AI/LLM-based tools for research acceleration, and JavaScript experience as a bonus. I bring all three. I have applied **Random Forest, XGBoost, CNN, PyTorch, and TensorFlow** to multiple translational genomics problems -- most directly, predicting biologic therapy response from **Janssen Phase III clinical trial data** integrating gene expression, genetic, clinical, and laboratory variables. I work with **Claude Code, Cursor, and Codex** daily for pipeline

development, and I have built production web-based tools with **JavaScript, HTML, and CSS** -- including a computer vision clinical tool for automated PASI scoring of psoriasis severity from sequential patient images. I understand how to communicate ML model outputs to both computational and non-computational audiences, which I do routinely with dermatologists, immunologists, and pharmaceutical partners.

CMC and Manufacturing Analytics

Kenai's role specifically requires partnering with manufacturing and CMC teams -- a context I understand from translational work. At UCSF I have collaborated directly with **Amgen, Johnson and Johnson, LEO Pharma, and TrexBio** to align computational analyses with therapeutic development priorities, delivering actionable insights within program timelines. I have contributed key analytical support to multiple NIH R01 grant applications and pharmaceutical research programs, always translating complex omics findings into clear narratives for cross-functional audiences including clinicians and program leadership. The discipline of communicating findings at the right level of abstraction for the right audience -- a core competency for a small, fast-moving organization like Kenai -- is something I have practiced extensively.

Scalable Pipeline Development and Data Infrastructure

I have built production bioinformatics pipelines using **Python, R, Bash, Snakemake, Docker, and AWS** for large-scale genomic data movement, metadata management, and multi-omics integration across HPC and cloud environments. I have designed complete data processing stacks from raw sequencing data through biological interpretation -- including a full GWAS pipeline across >20,000 samples and PRS scoring engines for polygenic risk quantification. I understand the importance of reproducibility, traceability, and long-term data architecture -- the infrastructure requirements your job description specifically highlights -- and I build with those principles from the start.

Neuroscience and Cell Therapy Alignment

I was particularly drawn to Kenai's focus on neurological disease. My UCSF work has included analysis of **neuronal compartment remodeling** in inflammatory tissue using Xenium Spatial Transcriptomics -- specifically the identification of MPZ+S100B+ Schwann cells as a disease-relevant population in atopic dermatitis, a finding with direct mechanistic parallels to neuroimmune interactions relevant to your programs. While my primary training is in skin immunology and computational genomics, the analytical frameworks for characterizing cellular heterogeneity, tracking cell states, and identifying therapeutic targets transfer directly to allogeneic cell therapy for neurological diseases. I am genuinely excited about applying these skills to a new disease area at the frontier of cell therapy.

My full publication record is available on [Google Scholar](#). My supervisor, **Professor Wilson Liao, M.D.** (Professor and Vice Chair of Research, Department of Dermatology, UCSF) has reviewed this application and expressed his strong support. He is available to speak directly to the quality, independence, and impact of my work and may be reached at wilson.liao@ucsf.edu. I am enthusiastic about contributing to Kenai's mission and welcome the opportunity to discuss how my background maps to your needs. Please reach me at sugandh.kumar@ucsf.edu or +1-628-246-6846 at any time. Thank you for your consideration.

Sincerely,

Sugandh Kumar, M.Sc., Ph.D.

Postdoctoral Research Fellow, UCSF | Bay Area, CA

Work Authorization: H1B (transferable) -- open to employer sponsorship